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CMPSC 132 – Final Project Report

Github: <https://github.com/rh38/laptopscraper>

Newegg Top-Seller Laptop Page Scraper

Description:

This project is a web scraper that pulls from the “Best-Seller” laptop page on the popular tech site Newegg.com. It sends a get request to the page, parses the html, and then displays each laptop. This includes the name price and rating if available. It handles errors in fetching parsing and more.

Approach and Logic:

My approach was first to find a good site. Once I did that, I used chrome dev tools to read the body and headers of the request to make my own request in python. I tried amazon but it seems like they didn't want people to scrape their site. I then tried Newegg and it worked great. Once I got a response, I then used ctrl + F to find keywords on the page like names and prices to locate the tabs. Once I got these tags, I could automate this process for all the laptops. The program is broken down into 4 main functions.

Get_page(url) – this fetches the page and returns the content

Save_page(html) – this was more for debugging and finding tags so not needed anymore

Find_laptops(html) - this is the function that parses the html into names, prices, and rating

Display_laptops(laptops) – this function takes the dicts returned by Find_laptops and displays them nicely

Data Structures Used:

Dictionaries

- Each laptop is stored as a dictionary with its three value keys
- The header for the request is also a dictionary

Lists

- All laptops' dictionaries are stored in a list

Challenges Faced:

[1]

My most annoying challenge was that in the headers of my request I originally was accepting encoding. It looked like "Accept-Encoding": "gzip, etc". When I was accepting encoding it returned garbage when I tried to print it. This took a minute to debug.

[2]

Finding the tags was also challenging because the page has a lot of data. Luckily Ctrl + F came in handy. It still took awhile to debug and get everything working.

Possible Improvements:

- Letting the user input a max price
- Letting the user choose how many to print
- Let user sort by rating or price
- Scrape more pages not just the first one